

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 4

9 Semantics of Programming Languages (nk480)

Consider the following small expression language:

Expressions	$e ::= x \mid \text{let } x = e \text{ in } e' \mid \underline{n} \mid e + e'$
	$\mid \text{true} \mid \text{false} \mid \text{if } e \text{ then } e' \text{ else } e''$
Values	$v ::= \underline{n} \mid \text{true} \mid \text{false}$
Evaluation Contexts	$E ::= \text{let } x = \square \text{ in } e' \mid \square + e \mid v + \square$
	$\mid \text{if } \square \text{ then } e' \text{ else } e''$
Environments	$\rho ::= \cdot \mid \rho, x = v$

with the following reduction semantics:

$$\begin{array}{c}
 \frac{\langle \rho; e \rangle \mapsto \langle \rho'; e' \rangle}{\langle \rho; E[e] \rangle \mapsto \langle \rho'; E[e'] \rangle} \qquad \frac{x \notin \text{dom}(\rho_1)}{\langle \rho_0, x = v, \rho_1; x \rangle \mapsto \langle \rho_0, x = v, \rho_1; v \rangle} \\
 \\
 \frac{}{\langle \rho; \underline{n}_1 + \underline{n}_2 \rangle \mapsto \langle \rho; \underline{n}_1 + \underline{n}_2 \rangle} \qquad \frac{}{\langle \rho; \text{let } x = v \text{ in } e \rangle \mapsto \langle \rho, x = v; e \rangle} \\
 \\
 \frac{}{\langle \rho; \text{if true then } e' \text{ else } e'' \rangle \mapsto \langle \rho; e' \rangle} \qquad \frac{}{\langle \rho; \text{if false then } e' \text{ else } e'' \rangle \mapsto \langle \rho; e'' \rangle}
 \end{array}$$

- (a) Give a set of typing rules for this language. [4 marks]

Answer: This is basically bookwork. First, we define types and contexts:

$$\begin{array}{l}
 \tau ::= \text{int} \mid \text{bool} \\
 \Gamma ::= \cdot \mid \Gamma, x : \tau
 \end{array}$$

Now, we can give the typing rules:

$$\begin{array}{c}
 \frac{}{\Gamma \vdash \text{true} : \text{bool}} \qquad \frac{}{\Gamma \vdash \text{false} : \text{bool}} \\
 \\
 \frac{}{\Gamma \vdash \underline{n} : \text{int}} \qquad \frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau} \\
 \\
 \frac{\Gamma \vdash e : \text{int} \quad \Gamma \vdash e' : \text{int}}{\Gamma \vdash e + e' : \text{int}} \qquad \frac{\Gamma \vdash e : \text{bool} \quad \Gamma \vdash e' : \tau \quad \Gamma \vdash e'' : \tau}{\Gamma \vdash \text{if } e \text{ then } e' \text{ else } e'' : \tau} \\
 \\
 \frac{\Gamma \vdash e : \tau \quad \Gamma, x : \tau \vdash e' : \tau'}{\Gamma \vdash \text{let } x = e \text{ in } e' : \tau'}
 \end{array}$$

- (b) Find an example expression which exhibits an anomalous execution behaviour, and briefly explain what the problem with the semantics is. [5 marks]

Answer: The reduction rule for $\text{let } x = v \text{ in } e$ adds a binding for x to ρ , and never discards it. As a result, the bindings for variables which have gone out of scope continue to shadow earlier bindings forever. This permits computing wrong values. Consider the expression:

$$\text{let } x = \underline{1} \text{ in } (\text{let } x = \underline{10} \text{ in } x) + x$$

— *Solution notes* —

This evaluates incorrectly to 20, rather than to 11.

- (c) Modify the semantics to repair the problem, and briefly explain how your solution works. [5 marks]

Answer: The most direct solution is to just rename the variables so that shadowing never arises:

$$\frac{y \notin \text{dom}(\rho)}{\langle \rho; \text{let } x = v \text{ in } e \rangle \mapsto \langle \rho, y = v; [y/x]e \rangle}$$

A more complicated (and efficient) solution would be to add rules to pop the binding for x after it goes out of scope.

- (d) Formulate progress and type preservation lemmas for the repaired language. (A proof is not necessary.) [6 marks]

Answer: The key to this question is to define a judgement $\vdash \rho : \Gamma$ for well-typed environments:

$$\frac{}{\vdash \cdot : \cdot} \qquad \frac{\vdash \rho : \Gamma \quad \cdot \vdash v : \tau}{\vdash \rho, x = v : \Gamma, x : \tau}$$

With this, we can now state progress and type preservation:

- (i) If $\Gamma \vdash e : \tau$ and $\vdash \rho : \Gamma$, then either e is a value, or there exist ρ' and e' such that $\langle \rho; e \rangle \mapsto \langle \rho'; e' \rangle$.
 - (ii) If $\Gamma \vdash e : \tau$ and $\vdash \rho : \Gamma$ and $\langle \rho; e \rangle \mapsto \langle \rho'; e' \rangle$, then there exists a $\Gamma' \supseteq \Gamma$ such that $\vdash \rho' : \Gamma'$ and $\Gamma' \vdash e' : \tau$.
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